Project: Worked examples
Title: Snap-fits: cantilever, annular, torsion
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1. Scope and method

A plastic snap-fit joint of one of five forms: a cantilever lug of constant section, of tapered thickness or of tapered width, an annular (closed ring) snap, or a torsion bar. The strain at the critical section against the permissible strain of the plastic (one-time or repeated joining), the deflection force with the secant modulus, the assembly force (inclined entry face) and the separation force (return face) with Coulomb friction, and whether the joint is self-locking.

WHAT IT CALCULATES	RELATION
Cantilever geometry (Euler–Bernoulli)	$y = K_y \cdot \varepsilon \cdot l^2/h$ with $K_y = 2/3$ (constant section; the design guides print the rounded 0.67); a linear taper of the thickness to $r_h = h_{tip}/h$ gives $K_y = 2 \cdot J/\varphi_{max}$ with $J = [2r_h - r_h^2/2 - \ln r_h - 3/2]/(1 - r_h)^3$ and $\varphi_{max} = 1$ for $r_h \geq 1/2$ (otherwise the strain peaks inside the beam at $t = (2a - 1)/a$, $a = 1 - r_h$, with $\varphi_{max} = (1 - t)/(1 - a \cdot t)^2$); a linear taper of the width to r_b gives $K_y = 2$...
Strain and force of the cantilever	$\varepsilon = y \cdot h/(K_y \cdot l^2)$; $P = b \cdot h^2 \cdot Es \cdot \varepsilon/(6 \cdot l \cdot \varphi_{max}) = (b \cdot h^2/6) \cdot (Es \cdot \varepsilon/l)$ at the root of a constant section; Es is the secant modulus at ε (typed, or interpolated from the W3 points Es(ε))
Annular (derived, free thin ring)	nominal hoop strain $\varepsilon_{nom} = \Delta D/D$ (the mean diameter D widened by the diametral interference ΔD); the strain checked is the larger of $\varepsilon = K_{ann} \cdot \varepsilon_{nom}$ (K_{ann} the user's strain factor) and the inner-surface strain $\Delta D/(D - h)$; the radial contact force summed round the ring is $P = K_F \cdot 2\pi \cdot Es \cdot \varepsilon_{nom} \cdot h \cdot b$ (hoop force $Es \cdot \varepsilon_{nom} \cdot h \cdot b$ over the radius, K_F the user's optional force factor, 1 if none), b being the...
Torsion bar of radius r with a lever of length a deflected y	$\varphi = y/a$; the bar is held at one end (free length $l_{eff} = l$) or at both ends with the lever in the middle ($l_{eff} = l/2$, two halves in parallel); $\gamma = \varphi \cdot r/l_{eff}$, equivalent tensile strain $\varepsilon = \gamma/(1 + \nu)$, $\gamma_{perm} = \varepsilon_{perm} \cdot (1 + \nu)$; $G = Es/(2(1 + \nu))$, $T = n \cdot G \cdot (\pi r^4/2) \cdot \varphi/l_{eff}$ with n = 1 or 2, $P = T/a$ (both ends: $P = 4 \cdot G \cdot J \cdot \varphi/(l \cdot a)$)
Joining forces	$W = P \cdot (\mu + \tan\alpha)/(1 - \mu \cdot \tan\alpha)$ on the entry face (angle α to the joining direction) and $W' = P \cdot (\mu + \tan\alpha')/(1 - \mu \cdot \tan\alpha')$ on the return face (α′); the joint is self-locking, with an infinite separation force, when $\alpha' \geq \arctan(1/\mu)$; assembly jams when $\mu \cdot \tan\alpha \geq 1$
Utilisation $u = \varepsilon/\varepsilon_{perm}$ (FAIL above 1); permissible deflection $y_{perm} = K_y \cdot \varepsilon_{perm} \cdot l^2/h$ (annular $\Delta D_{perm} = \varepsilon_{perm} \cdot \min(D/K_{ann}, D - h)$; torsion $y_{perm} = \gamma_{perm} \cdot l_{eff} \cdot a/r$)	

Basis: Cantilever, annular and torsion snap-fit relations[1](Section B); Tip deflection and surface strain of a tapered cantilever, by

integration[2].

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Form of snap-fit	Cantilever, tapered thickness		drawing: ENC-310-07 rev C	any
l	Length of the beam (cantilever); of the torsion bar: its free length when held at one end, or its whole length between the two supports when held at both ends with the lever in the middle	18	mm	drawing: ENC-310-07 rev C	2 … 100 mm
h	Thickness at the root (cantilever); wall thickness of the ring (annular)	2.2	mm	drawing: ENC-310-07 rev C	0.3 … 10 mm
b	Width at the root (cantilever); effective axial length of the ring (annular: that of a free ring, not the lip of a longer tube)	7	mm	drawing: ENC-310-07 rev C	1 … 50 mm
h_{ratio}	Tip thickness / root thickness (tapered thickness)	0.5		drawing: ENC-310-07 rev C	0.25 … 1
y	Deflection at the tip (cantilever) or at the end of the lever (torsion): the undercut to be cleared	1.2	mm	drawing: ENC-310-07 rev C	0.05 … 20 mm
α	Entry (assembly) angle α of the inclined face to the joining direction	30	°	drawing: ENC-310-07 rev C	5 … 60 °
α_r	Return (separation) angle α′ of the retaining face (90° is a face square to the joining direction)	65	°	drawing: ENC-310-07 rev C	10 … 90 °
μ	Coefficient of friction between the two parts (record snapfit.friction; no default: the pair of materials, from your reference or a W3 friction pair)	0.35		assumption: Example friction coefficient for a plastic-on-plastic pair. Enter μ for your pair of materials and surfaces from your reference or a test (record snapfit.friction).	0.05 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
Es_{source}	Where the secant modulus comes from: typed at the design strain, or interpolated from the Es(ε) points of a W3 material	Typed secant modulus		assumption: Secant modulus typed for this example; a W3 material with its Es(ε) points can be applied instead.	any
Es	Secant modulus Es at the design strain and temperature (annular: at the nominal hoop strain ΔD/D) (typed; no default: the plastic's stress–strain data)	2000	MPa	assumption: Example secant modulus at the design strain and temperature. Enter Es for your plastic from its stress–strain data or a W3 material; no plastics data are shipped.	10 ... 20000 MPa
loading	How often the joint is made: selects the one-time or the repeated permissible strain	One-time (assembled once or rarely)		spec: Enclosure ENC-310 §6: the lid is opened for service only	any
$eps_{perm,once}$	Permissible strain for one-time joining, as a fraction (0.04 is 4 %) at the temperature of use (no default: from the plastic's data)	0.04		assumption: Example permissible strain for one-time joining. Enter it for your plastic at the temperature of use from its data or a W3 material; no plastics data are shipped.	0.001 ... 0.3
T_{use}	Temperature of use. Recorded: Es and the permissible strain must be the values at this temperature	23	°C	spec: Enclosure ENC-310 §6: room temperature	-40 ... 120 °C

3. Assumptions

A1. Beam theory (Euler–Bernoulli) for the cantilever: a rigid root, a linear taper, a constant secant modulus Es at the design strain over the whole beam, small deflection. The strain is the nominal surface strain at the critical section, with no local concentration from the root fillet; the geometry factors are derived by integration (the 0.67 of the design guides is the constant-section 2/3 rounded). (Euler–Bernoulli beam theory (derived); the guide's closed forms for the constant section are to be checked against the owner's copy.)
A2. Es and the permissible strain are properties of the plastic at the temperature of use, with the loading (one-time or repeated) and the moisture state of the part. No values are supplied: they come from the plastic maker's data or a W3 material, and the temperature entered records what they were taken at. (No plastics data are shipped.)
A3. Annular snap: a free, thin, closed ring of effective axial length b. The nominal hoop strain is the diametral interference over the mean diameter, ΔD/D; the strain checked is the larger of that times the user's strain factor K_{ann} (record snapfit.annular-factor, no default) and the inner-surface strain. The contact force is the hoop force $Es \cdot \varepsilon_{nom} \cdot h \cdot b$ taken round the ring, from the nominal strain only, times the

user's optional force factor K_F (record snapfit.annular-force-factor; blank is 1); a tube behind the ring stiffens the joint and raises the force, which only K_F can represent. The strain on the inner (contact) surface is ΔD/(D − h), slightly above the mean-diameter value, so the check takes the larger of the two; a wall above a tenth of the diameter warns that the thin-ring model has run out. (Derived from ring geometry (not a relation of the design guide); compare with the owner's copy of the guide.)
A4. Torsion bar: a solid round bar of radius r loaded at the end of a lever, the angle small (φ = y/a), held at one end or at both ends with the lever in the middle (twice the strain and four times the stiffness of one end for the same total length). The permissible shear strain is $\varepsilon_{perm}\cdot(1 + \nu)$ and the equivalent tensile strain $\gamma/(1 + \nu)$; G = Es/(2(1 + ν)) with the secant modulus at that strain. ($\gamma = \varphi\cdot r/l$ and $\gamma_{perm} \approx \varepsilon_{perm}\cdot(1 + \nu)$ are the brief's design-guide relations, to be checked against the owner's copy; the torque, the force and the two-ended case are derived.)
A5. Assembly and separation: a rigid mating part, Coulomb friction μ on the inclined face (α for assembly, α′ for separation), the deflection force P taken at the design deflection; no creep or stress relaxation of the joint in service, no impact (speed) effect. (Friction on an inclined plane; the friction coefficient is the user's record.)
A6. μ (Coefficient of friction between the two parts (record snapfit.friction; no default: the pair of materials, from your reference or a W3 friction pair)) is taken as 0.35. (Example friction coefficient for a plastic-on-plastic pair. Enter μ for your pair of materials and surfaces from your reference or a test (record snapfit.friction).)
A7. Es_{source} (Where the secant modulus comes from: typed at the design strain, or interpolated from the Es(ε) points of a W3 material) is taken as Typed secant modulus. (Secant modulus typed for this example; a W3 material with its Es(ε) points can be applied instead.)
A8. Es (Secant modulus Es at the design strain and temperature (annular: at the nominal hoop strain ΔD/D) (typed; no default: the plastic's stress–strain data)) is taken as 2000 MPa. (Example secant modulus at the design strain and temperature. Enter Es for your plastic from its stress–strain data or a W3 material; no plastics data are shipped.)
A9. $eps_{perm,once}$ (Permissible strain for one-time joining, as a fraction (0.04 is 4 %) at the temperature of use (no default: from the plastic's data)) is taken as 0.04. (Example permissible strain for one-time joining. Enter it for your plastic at the temperature of use from its data or a W3 material; no plastics data are shipped.)
A10. Es and the permissible strain are the values of the plastic at the temperature of use, 23 °C, as entered, in the moisture state of the part; the sheet cannot check this. (Recorded temperature of use.)

4. Working

[10] l_h = l / h
= 18 mm / 2.2 mm
= 8.1818
Slenderness of the beam l/h

[12] $J = [2r - r^2/2 - \ln r - 3/2]/(1 - r)^3$ ²((derived))
= r = 0.5
= 0.54518
Deflection integral J of the thickness taper (derived: Euler–Bernoulli integration)

[13] $\varphi_{max} = (1 - t^*)/(1 - \alpha\cdot t^*)^2$, $t^* = (2\alpha - 1)/\alpha$ for α > 1/2, else 1 ²((derived))
= α = 0.5
= 1
Largest value of the strain shape along the beam (1 at the root)

[14] $K_y = 2\cdot J/\varphi_{max}$ ²((derived))
= 2 × 0.54518 / 1
= 1.0904
Deflection factor: $y = K_y\cdot\varepsilon\cdot l^2/h$

[15] eps = y * h / (K_y * l^2) ²((derived))
= 1.2 mm × 2.2 mm / (1.0904 × (18 mm)^2)
= 0.74729 %

Maximum surface strain at the critical section (nominal)

[18] $eps_{perm} = eps_{perm,once}$ [1](Plastics: A Design Guide)

$= 0.04$

$= 4\ \%$

Permissible strain (one-time joining, as entered)

[21] $P = b * h^2 * Es * eps / (6 * l * \varphi_{max})$ [2]((derived))

$= 7\ mm \times (2.2\ mm)^2 \times 2000\ MPa \times 0.74729\ \% / (6 \times 18\ mm \times 1)$

$= 4.6886\ N$

Deflection force at the tip: P = (b·h²/6)·(Es·ε/l) at the root of a constant section

[22] $y_{perm} = K_y * eps_{perm} * l^2 / h$ [2]((derived))

$= 1.0904 \times 4\ \% \times (18\ mm)^2 / 2.2\ mm$

$= 6.4232\ mm$

Permissible tip deflection

[23] $strain_{util} = eps / eps_{perm}$

$= 0.74729\ \% / 4\ \%$

$= 0.18682$

Strain / permissible strain

[24] $tan_{\alpha} = tan(\alpha)$

$= tan(30\ deg)$

$= 0.57735$

tan α of the entry face

[25] $\mu_{tan,\alpha} = \mu * tan_{\alpha}$

$= 0.35 \times 0.57735$

$= 0.20207$

μ·tan α of the entry face

[26] $assembly_{force} = P * (\mu + tan_{\alpha}) / (1 - \mu_{tan,\alpha})$ [1](Plastics: A Design Guide)

$= 4.6886\ N \times (0.35 + 0.57735) / (1 - 0.20207)$

$= 5.4491\ N$

Assembly force W = P·(μ + tan α)/(1 − μ·tan α)

[27] $\alpha_{lock} = arctan(1/\mu)$ [1](Plastics: A Design Guide)

$= arctan(1/0.35)$

$= 70.71\ deg$

Return angle above which the joint is self-locking (π/2 when μ = 0)

[28] $tan_{ar} = tan(\alpha_r)$

$= tan(65\ deg)$

$= 2.1445$

tan α′ of the return face

[29] $retention_{force} = P * (\mu + tan_{ar}) / (1 - \mu * tan_{ar})$ [1](Plastics: A Design Guide)

$= 4.6886\ N \times (0.35 + 2.1445) / (1 - 0.35 \times 2.1445)$

$= 46.891\ N$

Separation force W′ = P·(μ + tan α′)/(1 − μ·tan α′)

5. Results

SYMBOL	VALUE	DESCRIPTION
eps	0.74729 %	Maximum strain
eps_{perm}	4 %	Permissible strain
$strain_{util}$	0.18682	Strain / permissible strain
Es	2000 MPa	Secant modulus used
P	4.6886 N	Deflection force
y_{perm}	6.4232 mm	Permissible deflection
l_h	8.1818	Slenderness l/h

SYMBOL	VALUE	DESCRIPTION
K_y	1.0904	Deflection factor K_y
φ_{max}	1	Strain shape maximum
$assembly_{force}$	5.4491 N	Assembly force
α_{lock}	70.71 deg	Self-locking return angle
$retention_{force}$	46.891 N	Separation force
$self_{locking}$	"No"	Self-locking
$fillet_{advisory}$	"The strain above is the nominal strain, with no local concentration. A sharp internal corner at the root (or at the end of the bar or ring) concentrates the strain and is the usual place for a snap-fit to break: give the root a generous fillet radius, of the order of the thickness, taper the section away from it, and apply a concentration factor from your own reference if the radius is small."	Root-fillet advisory

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Strain at the critical section against the permissible strain	$strain_{util}$ = 0.18682	$strain_{util}$ ≤ 1	0.187	[1](Plastics: A Design Guide)
PASS	Assembly: the entry face does not jam (μ·tan α < 1)	μ·tan α = 0.20207	< 1	0	[1](Plastics: A Design Guide)

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: $eps_{perm,once}$ (Material properties); μ (Coefficient of friction); Es (Secant modulus at the design strain). Enter each as one so its source, edition and licence travel with it.

8. Verification

Template mech.snap-fit@1.3.1 Content hash 0d39506ffeadbb9c Engine calc-engine/1.0.0
Known-answer cases passed 11/11 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Covestro, Snap-Fit Joints for Plastics: A Design Guide.
2. Euler–Bernoulli beam theory.